Abdul Basit Tonmoy

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EDUCATION

Wabash College — Crawfordsville, IN
Bachelor of Arts in Computer Science and Mathematics with Minor in Physics, GPA: 3.6/4.0 — *May 2027*
Relevant Coursework: Data Structures & Algorithms, Machine Organization, OOP, Machine Learning, Data Science, Numerical Methods, Functional Programming, Stochastic Simulation, Multivariable Calc, Linear Algebra, Autonomous Robotics

RESEARCH INTERESTS

Efficient Multimodal Representation Learning, Video Understanding, Vision-Language Models, Cross-Domain Transfer (Vision → Biology), Scalable AI Systems

ACHIEVEMENTS

- **Published Research (3rd Author):** Powers, G., Jin, A., **Tonmoy, A.B.**, Cao, H., Gohel, H., & Deng, Q. (2025). *Advanced Facial Emotion Classification with 135 Classes for Enhanced Cybersecurity Applications.* IEEE ICAIC 2025, Houston, TX
- **Research Manuscript in Preparation (1st Author):** *HMMD: Hierarchical Multi-Modal Deduplication with Adaptive Density Selection for VLM Inference*
- **J. Crawford Polley Mathematical Writing Prize** (May 2025) – Awarded for excellence in mathematical exposition and technical writing
- **Dean's List | Hackathon Wins**: CalHacks-12 (World's Largest, UC Berkeley), UAP CyberSiege-25

TECHNICAL SKILLS

Languages:	Python, C/C++, JavaScript, SQL, HTML/CSS
ML/AI Frameworks:	PyTorch, scikit-learn, Transformers, LangChain, HuggingFace
Data Science:	pandas, NumPy, SciPy, Matplotlib, Apache Airflow
Research Tools:	OpenCV, CLIP, ProtT5-Large, ESM-Fold, NLP (BERT, NER), ChromaDB
Cloud & DevOps:	GCP, AWS, Docker, Git, MongoDB

RESEARCH EXPERIENCE

Department of Mathematics and Computer Science, Wabash College — Crawfordsville, IN
Undergraduate Research Assistant - Autonomous Robotics (Advisor: Prof. Qixin Deng) — *May 2025 – Aug. 2025*
- Conducted funded research on multi-sensor fusion for autonomous navigation, integrating GPS, computer vision, and LiDAR to learn unified representations from heterogeneous data streams
- Designed real-time control algorithms achieving 97% obstacle detection accuracy and 2-meter positional precision, demonstrating efficient inference under computational constraints
- Developed adaptive fail-safe mechanisms validated across varying environmental conditions (lighting, weather, terrain), building systems that respond to context rather than fixed rules
- Presented findings to 60+ interdisciplinary audience, translating technical concepts for diverse backgrounds

RESEARCH & TECHNICAL PROJECTS

HMMD with Adaptive Density Selection for Vision Language Models — Oct. 2025 – Present
Python, CLIP, LPIPS, pHash/dHash/wHash, PySceneDetect, Whisper, OpenCV
- Developed novel seven-stage pipeline for efficient VLM inference, combining hierarchical deduplication with importance-weighted Non-Maximum Suppression for frame selection
- Designed three-tier cascade (hash voting ensemble, LPIPS, CLIP embeddings) ordered by computational cost, with adaptive density allocation based on scene complexity
- Achieved 84.5% average frame reduction (0% to 96.8% range) and 98.3% API cost savings on 1706 diverse advertisements while maintaining 100% extraction success rate
- Demonstrated content-aware adaptability: preserving frames when genuinely diverse, aggressively deduplicating static content, addressing core efficiency challenge in video understanding

HackEnzyme: Generative Enzyme Design from Molecular Structures (HackHarvard) — Oct. 2024
Python, PyTorch, ProtT5-Large, ESM-Fold, Transformers, HuggingFace
- Built end-to-end pipeline generating enzyme sequences from SMILES molecular representations, fine-tuning ProtT5-Large on custom dataset scraped from bioinformatics APIs
- Integrated ESM-Fold for 3D protein structure visualization, enabling interpretable outputs from sequence predictions
- Explored how representation learning transfers between vision and biological sequence domains

Advanced Emotion Recognition for Cybersecurity (Published, IEEE ICAIC) — Jan. 2024 – May 2025
Python, PyTorch, U-Net, ResNet
- Co-architected hybrid framework combining U-Net (spatial attention) and ResNet (feature extraction) to classify 135 fine-grained emotion classes, co-authoring the comprehensive technical methodology section detailing the model's design
- Contributed to theoretical analysis in the discussion section, synthesizing performance evaluations against SOTA benchmarks and addressing challenges of annotation inconsistency in high-dimensional affective computing

Formula 1 Predictive Analytics System — June 2025
Python, SQL, LightGBM, AWS, Apache Airflow
- Built end-to-end ML pipeline processing 50,000+ telemetry records, achieving 87% race outcome prediction through feature engineering on high-dimensional sensor data
- Engineered cloud infrastructure with automated ETL workflows, demonstrating production-scale systems thinking

PROFESSIONAL EXPERIENCE

Auxiliary Services, Wabash College Crawfordsville, IN
Data Analyst *Jan. 2024 – May 2025*

- Designed automated pipeline processing 15+ years of heterogeneous records, delivering $20,160 annual savings through data-driven recommendations
- Transformed unstructured data into unified analytical framework, demonstrating ability to extract signal from noisy, real-world datasets

Motorpool, Wabash College Crawfordsville, IN
Software Engineering Intern *May 2024 – Aug. 2024*

- Built RAG-based assistant using LangChain and GPT-3.5, reducing manual workload by 25% through natural language understanding
- Architected full-stack platform with 30% query optimization, designing systems that adapt to user needs

TEACHING & LEADERSHIP

Resident Advisor Crawfordsville, IN
Wabash College *Aug. 2024 – Present*

- Mentor 100+ student community, building environment where struggle is normalized and support is accessible
- Serve as first responder coordinating with campus security and administration, demonstrating ability to handle high-pressure situations with sound judgment

Teaching Assistant - Physics 109 & 110 Crawfordsville, IN
Department of Physics, Wabash College *Aug. 2024 – May 2025*

- Assisted with laboratory sessions and problem-solving workshops for introductory physics courses, providing one-on-one guidance to students on experimental techniques, data analysis, and conceptual understanding
- Graded assignments and lab reports, providing constructive feedback to support student learning and mastery of physics principles

Mathematics and Computer Science Mentor Crawfordsville, IN
Quantitative Skills Center, Wabash College *Jan. 2026 – Present*

- Provide peer mentorship across introductory to advanced coursework, making quantitative knowledge accessible

GRADUATE SCHOOL GOALS

Pursuing Ph.D. in Computer Science or Electrical Engineering focused on efficient multimodal representation learning. Goal: develop systems that bridge vision, language, and biological modalities while lowering computational barriers, ensuring powerful AI tools remain accessible beyond well-resourced institutions.